



PRODUCT NAME

High-performance Organic Shower Gel Formula

Description

An eco-friendly, cost-effective shower gel with excellent cleansing properties, pH balanced for skin compatibility, and formulated with biodegradable ingredients for minimal environmental impact.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100 kg to 500 kg
Processing Time:	4 hours
Temperature:	60°C to 40°C
Storage Conditions:	Store in a cool, dry place away from direct sunlight
Equipment:	Mixing vessel with heating and cooling capabilities, pH meter, homogenizer
Certification:	EcoCert, COSMOS

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	60.0%	Solvent
2	Sodium Coco-Sulfate	Sodium Coco-Sulfate	15.0%	Cleansing Agent
3	Cocamidopropyl Betaine	Cocamidopropyl Betaine	10.0%	Surfactant, Foam Booster
4	Glycerin	Glycerin	5.0%	Humectant
5	Citric Acid	Citric Acid	2.0%	pH Adjuster
6	Sodium Benzoate	Sodium Benzoate	0.5%	Preservative
7	Potassium Sorbate	Potassium Sorbate	0.5%	Preservative
8	Xanthan Gum	Xanthan Gum	2.0%	Thickener
9	Fragrance	Fragrance	0.5%	Scent
10	Sodium Chloride	Sodium Chloride	4.5%	Viscosity Adjuster
Total			100%	

Manufacturing Process

Phase 1: Phase A

1. Heat the Aqua to 60°C in a main kettle.
2. Dissolve the Sodium Coco-Sulfate and Cocamidopropyl Betaine into the warmed Aqua, stirring continuously.
3. Add Glycerin while maintaining the temperature and stir until homogeneous.

Phase 2: Phase B

1. Slowly sprinkle Xanthan Gum into the mixture while stirring to avoid clumping.
2. Ensure the Xanthan Gum is fully hydrated before proceeding to the next step.

Phase 3: Phase C

1. Lower the temperature to 40°C.
2. Add Citric Acid incrementally to adjust the pH to exactly 7, testing the pH frequently.
3. Incorporate Sodium Benzoate, Potassium Sorbate, and Fragrance into the batch and mix thoroughly.

Phase 4: Phase D

1. Add Sodium Chloride to achieve the desired medium viscosity.
2. Cool the mixture to room temperature, ensuring it remains well-mixed and homogeneous.

Required Equipment

Mixing vessel with heating and cooling capabilities, pH meter, homogenizer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: EcoCert, COSMOS

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100 kg to 500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a small amount of gel to wet skin, lather, and rinse thoroughly for a refreshing clean.